INCH-POUND

MIL-DTL-28840/9F
23 October 2009
SUPERSEDING
MIL-DTL-28840/9E
20 April 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, 45°, CABLE SEALING AND SHIELD TERMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

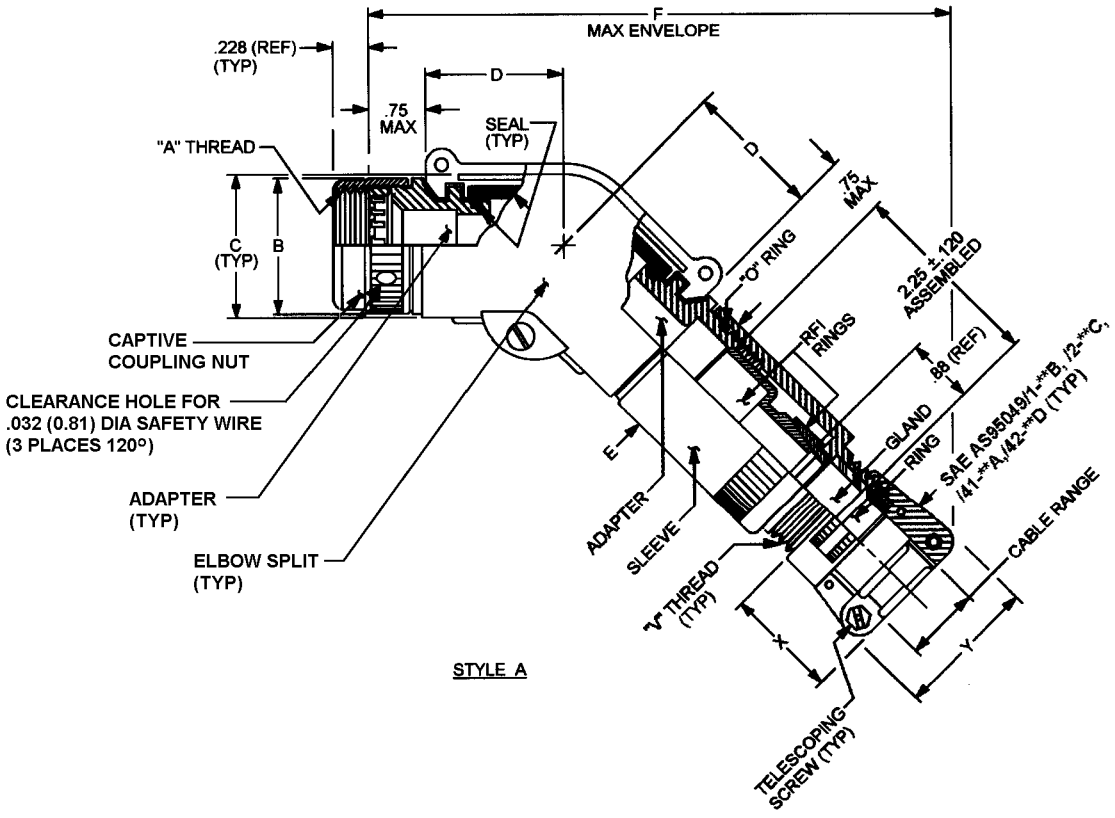
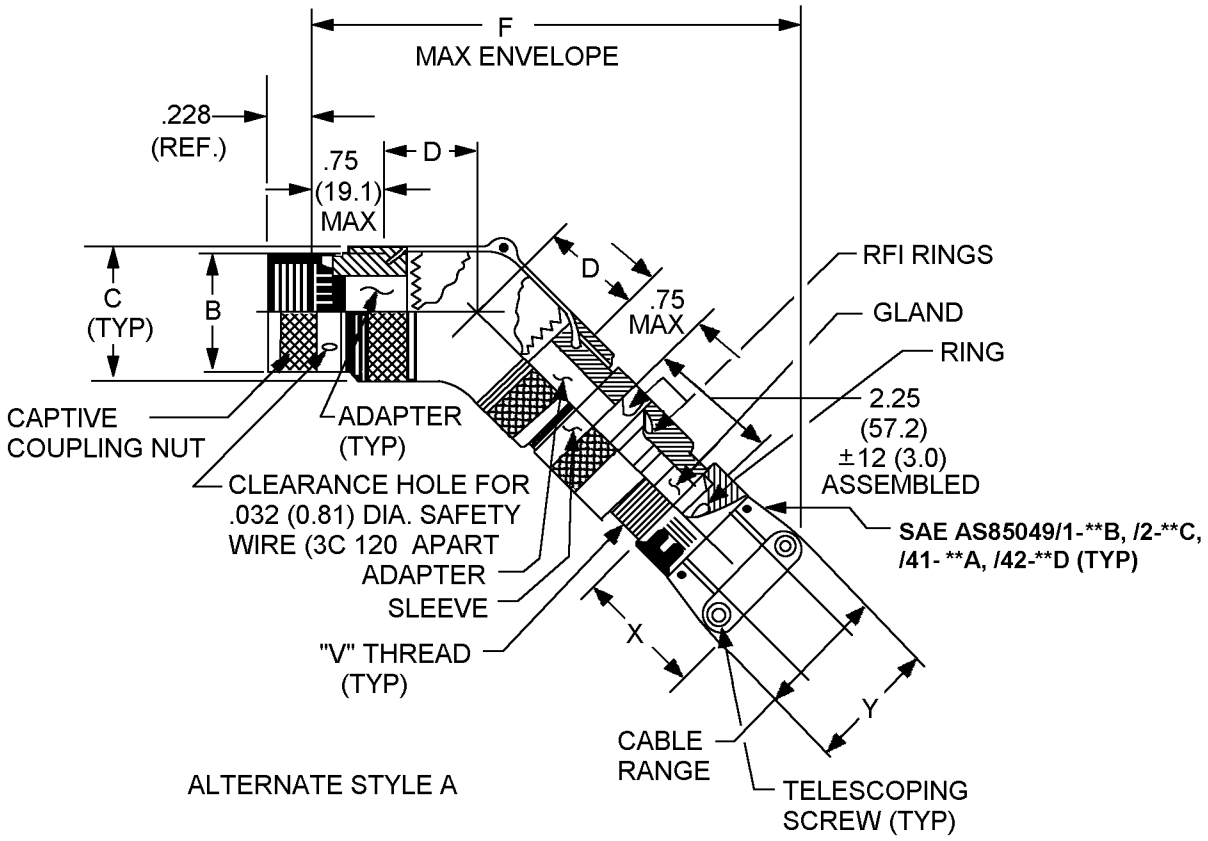


FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5935

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Inches	mm
.12	3.05
.228	5.79
.75	19.05
.88	22.35
2.25	57.15

FIGURE 1. Dimensions and configurations - Continued.

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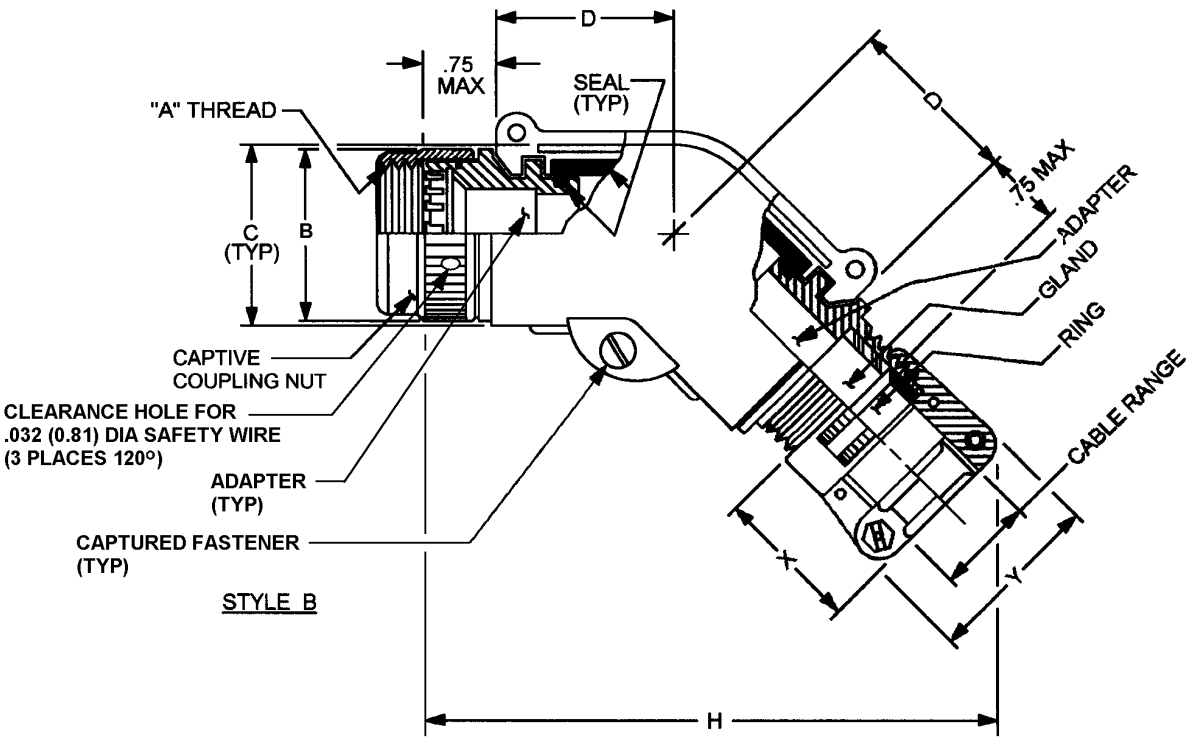


FIGURE 1. Dimensions and configurations - Continued.

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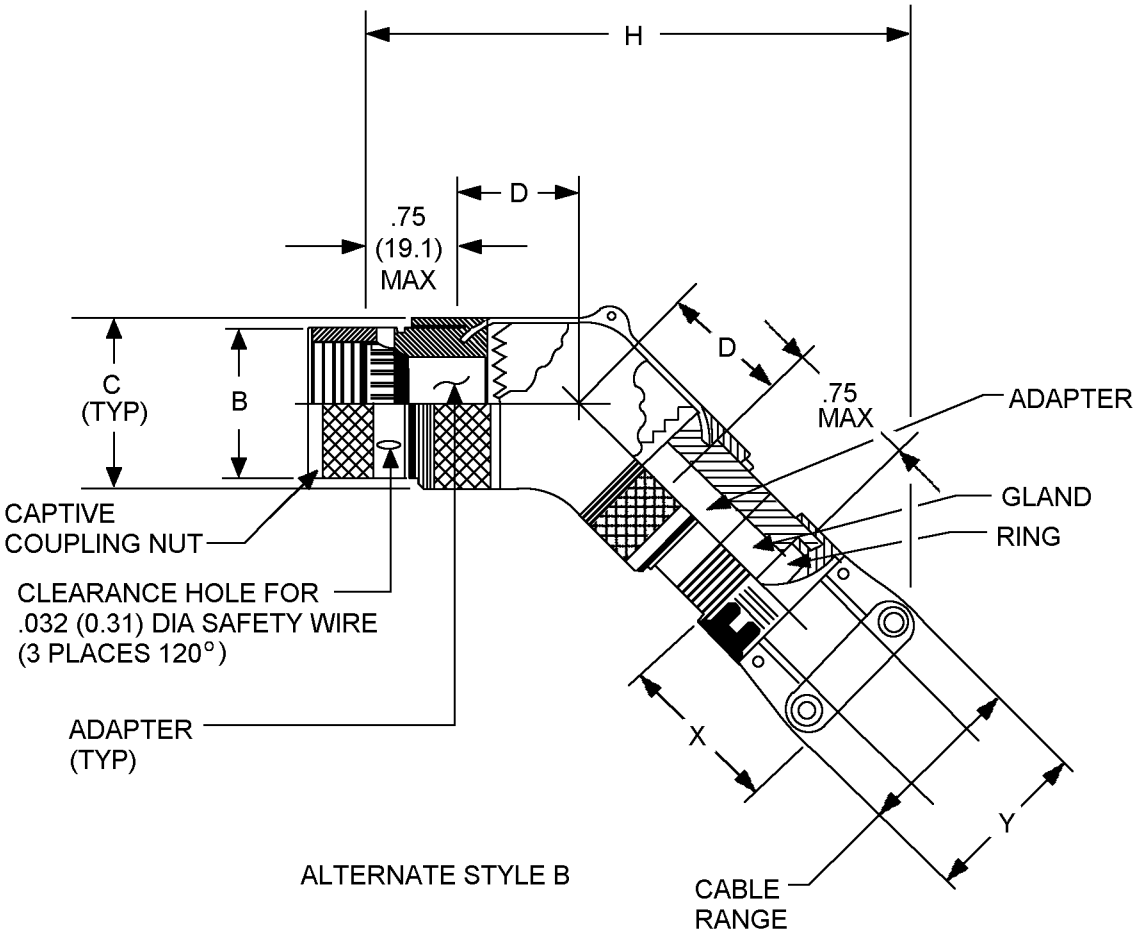


FIGURE 1. Dimensions and configurations - Continued.

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Backshell size (see note 3)	Dash no.	Designator (see note 5)	A thread	B max dia	C max dia	D max
A	01	A (11)	.750-20 UNEF	1.028 (26.11)	1.031 (26.19)	1.031 (26.19)
A	02	B (13)	.875-20 UNEF	1.141 (28.98)		
B	03	B (13)				
A	04	C (15)	1.00-20 UNEF	1.263 (32.08)	1.281 (32.54)	1.156 (29.36)
B	05	C (15)				
C	06	C (15)				
D	07	C (15)			1.513 (38.89)	1.281 (32.54)
A	08	D (17)	1.125-18 UNEF	1.387 (35.23)		
B	09	D (17)				
C	10	D (17)				
A	11	E (19)	1.250-18UNEF	1.513 (38.43)		
B	12	E (19)			1.781 (45.24)	1.406 (35.71)
A	13	F (23)	1.438-18 UNEF	1.703 (43.51)		
B	14	F (23)				
C	15	F (23)				
A	16	G (25)	1.562-18 UNEF	1.825 (46.35)		
B	17	G (25)				
C	18	G (25)			2.031 (51.59)	1.531 (38.89)
D	19	G (25)				
A	20	H (29)	1.875-16 UN	2.143 (54.43)		
B	21	H (29)				

FIGURE 1. Dimensions and configurations - Continued.

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Backshell size (see note 3)	Dash no.	Designator (see note 5)	Cable range		E max	F max style A	H max style B
			max	min			
A	01	A (11)	.312 (7.92)	.188 (4.77)	1.156 (29.36)	5.879 (149.33)	4.199 (106.65)
A	02	B (13)	.312 (7.92)	.188 (4.77)		5.879 (149.33)	4.199 (106.65)
B	03	B (13)	.438 (11.12)	.281 (7.14)		5.932 (150.67)	4.252 (108.00)
A	04	C (15)	.312 (7.92)	.188 (4.77)	1.281 (32.54)	6.092 (154.74)	4.412 (112.06)
B	05	C (15)	.438 (11.12)	.281 (7.14)		6.146 (156.11)	4.466 (113.44)
C	06	C (15)	.562 (14.27)	.344 (8.74)		6.202 (157.53)	4.522 (114.86)
D	07	C (15)	.750 (19.05)	.438 (11.12)	1.406 (35.71)	6.533 (165.94)	4.853 (123.27)
A	08	D (17)	.625 (15.87)	.375 (9.62)		6.413 (162.89)	4.733 (120.22)
B	09	D (17)	.750 (19.05)	.438 (11.12)		6.533 (165.94)	4.853 (123.27)
C	10	D (17)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	6.636 (168.55)	4.956 (125.88)
A	11	E (19)	.625 (15.87)	.375 (9.52)	1.406 (35.71)	6.413 (162.89)	4.733 (120.22)
B	12	E (19)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	6.846 (173.89)	5.166 (131.22)
A	13	F (23)	.750 (19.05)	.438 (11.12)	1.406 (25.71)	6.747 (171.37)	5.067 (128.70)
B	14	F (23)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	6.846 (173.89)	5.166 (131.22)
C	15	F (23)	1.388 (30.17)	.812 (20.62)	1.781 (45.24)	7.103 (180.42)	5.423 (138.23)
A	16	G (25)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	6.846 (173.89)	5.156(130.96)
B	17	G (25)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)	7.103 (180.42)	5.4239 (138.23)
C	18	G (25)	1.250 (31.75)	.875 (22.23)		7.316 (185.83)	5.636 (143.15)
D	19	G (25)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)	7.425 (188.60)	5.745 (145.92)
A	20	H (29)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)	7.316 (185.83)	5.636 (143.15)
B	21	H (29)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)	7.425 (188.60)	5.745 (145.92)

FIGURE 1. Dimensions and configurations - Continued.

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Backshell size (see note 3)	Dash no.	Designator (see note 5)	V thread class	X ref dim	Y ref dim
A	01	A (11)	.625-24 UNEF	1.281 (32.54)	.938 (23.82)
A	02	B (13)	.625-24 UNEF		.938 (23.82)
B	03	B (13)	.750-20 UNEF		1.125 (28.57)
A	04	C (15)	.625.24 UNEF		.938 (23.82)
B	05	C (15)	.750-20 UNEF		1.125 (28.57)
C	06	C (15)	.875-20 UNEF		1.312 (33.32)
D	07	C (15)	1.188-18 UNEF	1.312 (33.32)	1.593 (40.62)
A	08	D (17)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
B	09	D (17)	1.188-18 UNEF	1.312 (33.32)	1.395 (40.62)
C	10	D (17)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
A	11	E (19)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
B	12	E (19)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
A	13	F (23)	1.188-18 UNEF	1.312 (33.32)	1.593 (40.62)
B	14	F (23)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
C	15	F (23)	1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
A	16	G (25)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
B	17	G (25)	1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
C	18	G (25)	1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
D	19	G (25)	2.000-16 UNS	1.625 (41.28)	2.343 (59.51)
A	20	H (29)	1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
B	21	H (29)	2.000-18 UNS	1.625 (41.28)	2.343 (59.51)

FIGURE 1. Dimensions and configurations - Continued.

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Backshell size (see note 3)	Dash no.	Designator (see note 5)	A thread	B max dia	C max dia	D max
A	22	J (33)	2.062-16 UNEF	2.329 (59.16)	2.531 (64.29)	1.781 (45.24)
B	23	J (33)				
B	24	A (11)	.750-20 UNEF	1.028 (26.11)	1.031 (26.19)	1.031 (26.19)
C	25	A (11)			1.281 (32.54)	1.156 (29.36)
C	26	B (13)	.875-20 UNEF	1.141 (28.98)	1.281 (32.54)	1.156 (29.36)
D	27	B (13)			1.531 (38.89)	1.281 (32.54)
E	28	C (15)	1.000-20 UNEF	1.263 (32.08)		

Backshell size (see note 3)	Dash no.	Designator (see note 5)	Cable range		E max dim	F max style A	H max style B
			Max	Min			
A	22	J (33)	1.380 (35.05)	1.000 (25.40)	2.281 (57.94)	7.852 (199.44)	6.127 (156.77)
B	23	J (33)	1.625 (41.28)	1.250 (31.75)		6.454 (163.93)	6.454 (163.93)
B	24	A (11)	.438 (11.12)	.281 (7.14)	1.156 (29.36)	5.932 (150.67)	4.252 (108.00)
C	25	A (11)	.562 (14.27)	.344 (8.74)	1.281 (32.54)	6.202 (157.53)	4.522 (114.86)
C	26	B (13)	.562 (14.27)	.344 (8.74)	1.281 (32.54)	6.202 (157.53)	4.522 (114.86)
D	27	B (13)	.625 (15.87)	.375 (9.52)	1.406 (35.71)	6.413 (162.89)	4.733 (120.22)
E	28	C (15)	.625 (15.87)	.375 (9.50)		6.413 (162.89)	4.733 (120.22)

FIGURE 1. Dimensions and configurations - Continued.

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Backshell size (see note 3)	Dash no.	Designator (see note 5)	V thread class	X ref dim	Y ref dim
A	22	J (33)	2.000-18 UNS	1.625 (41.28)	2.343 (59.51)
B	23	J (33)	2.250-16 UNS	1.900 (48.26)	2.750 (69.85)
B	24	A (11)	.750-20 UNEF	1.281 (32.54)	1.125 (28.57)
C	25	A (11)	.875-20 UNEF	1.281 (32.54)	1.312 (33.32)
C	26	B (13)	.875-20 UNEF	1.281 (32.54)	1.312 (33.32)
D	27	B (13)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
E	28	C (15)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Designator of backshell size within each shell size.
4. Dimensions apply after plating.
5. Shell sizes are provided within parentheses for information and are not a part of the designator.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS

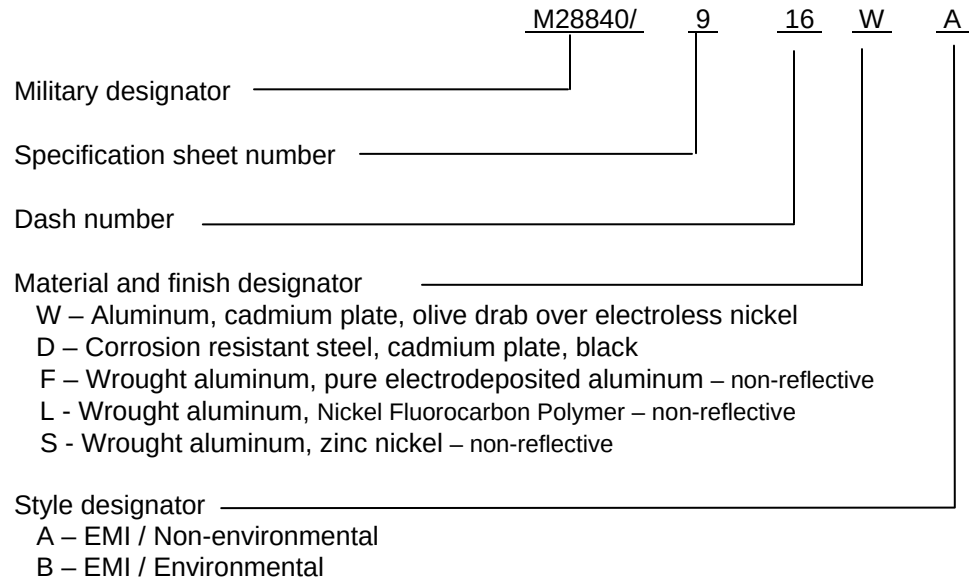
Dimensions and configurations: See figure 1.
Interface dimensions shall conform to MIL-DTL-28840.

Material: Shall be aluminum alloy in accordance with ASTM B85/B85M, ASTM B26/B26M, or ASTM B108/B108M. Corrosion resistant steel shall be in accordance with ASTM A276, ASTM A580/A580M or ASTM A473, type 316.

Finish: Aluminum components shall plated in accordance with MIL-DTL-28840. Corrosion resistant steel components shall be cadmium plate, black. The screws and washers shall be passivate in accordance with SAE-AMS2700, type 2.

Marking: The backshell shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

PIN example:



Qualification: Two samples in each shell size for which qualification is desired and shall be tested in accordance with MIL-DTL-28840.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Referenced documents. In addition to MIL-DTL-28840, this document references the following:

ASTM A276
ASTM A473
ASTM A580/A580M
ASTM B26/B26M
ASTM B85/B85M
ASTM B108/B108M
SAE-AMS2700

CONCLUDING MATERIAL

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2007-221)

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